

Antoine Lorient

RESEARCHER IN APPLIED MACHINE LEARNING FOR HUMAN AND COMPUTER INTERACTION

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Skills

Programming	Python, Embedded C, Javascript, SQL, LaTeX
Scientific Python	Numpy, Pandas, Scipy, Seaborn, Plotly, Dash
Machine Learning Python	Pytorch, Keras, Pytorch Lightning, Cuda, MIFlow, Scikit-Learn, Pyro
Research	Scientific Writing, Teaching, Tutoring, Quantitative and Qualitative analysis, Experiment Design, Ethics
Languages	French, English, Swedish

Research

CNRS - IRCAM - INRIA

Paris, France

POST-DOCTORAL RESEARCHER

Jan. 2020 - Sept. 2022

- Research with Pr. Bevilacqua on movement exploration and modeling, demoed at *NIME 2022*.
- Research with Dr. Caramiaux on the perception of motor learning in tasks involving complex and creative motions, such as dance or music practise, published in *PLOS ONE*.
- Deployed a metric learning approach to a in-house dataset including IMUs time series and videos recording, using DTW cython. Later remade with modern tools - pytorch and cuda.

University of Glasgow

Glasgow, United Kingdom

RESEARCH AND TEACHING ASSESSMENT ASSISTANT

Mar. 2019 - Sep. 2019

- Research with Pr. Chalmers on visualisation techniques for low-dimensional embeddings with a focus on bridging t-SNE and force-based approaches. Production of Jupyter notebooks gathering the Python implementation of proposed algorithms and results of experiments on Mnist-Fashion.
- Assessment of undergraduate projects and tutoring courses. Correction of examination essays.

University of Glasgow

Glasgow, United Kingdom

PHD CANDIDATE AND CONTRIBUTOR TO THE EUROPEAN PROJECT MOREGRASP

Mar. 2015 - Feb. 2019

- Member of the Inference, Dynamics and Interaction (IDI) group led by Pr. Murray-Smith.
- Research involving computational methods for the study of novel gestural interactions: use of discriminative models to create virtual surfaces from RGB-D sensors, optimisation of gameplay experiences for custom game controllers via user probabilistic modelling, audio-based reinforcement learning for the exploration and characterisation of user physical capabilities. Partly published at *ISS'17*.
- Contributions to the European project MoreGrasp: lead redaction of deliverable documents, implementation of visual feedback mechanisms on smartwatches, preparation of live demos and posters for review meetings, establishment of contacts with practitioners in the QEU hospital in Glasgow leading to workshops in-situ and the writing of a funding proposal targeting a local charity.
- Tutoring for Artificial Intelligence and Data Fundamentals courses (undergraduate, Master levels).

Mobile Vinn Excellence Center (Mobile Life)

Stockholm, Sweden

RESEARCH ASSISTANT

Feb. 2014 - Oct. 2014

- Member of Pr. Brown's group dedicated to the understanding of human interactions with novel technology through the use of ethnographic methods.
- Research on innovative designs for always-on in the background speech interactions. Design of an experiment involving day-long recordings of participants' everyday life. Qualitative analysis and design workshop with trained Interaction designers. Published at *CHI'15*.

Industry

Now & Net

Paris, France

PARTNER

2013 - 2015

- Member of a startup, created in 2012, aiming at providing a collective musical experience organised around DJ-ing.
- Writing of research and development plans leading to the successful application to the status of young innovative company (JEI). Development of a proof-of-concept recommender system based on the technology Neo4j, a graph-based database.

Ericsson AB

Stockholm, Sweden

MULTIPLE POSITIONS

2008 - 2013

- SCRUM master and Software Developer - 2013/2014: Acting as the “guardian of the process” to facilitate daily operations of a cross functional team (8 people). Animation of planning meetings, daily follow-up meetings and retrospective meetings.
- System engineer at AT&T laboratory - summer 2012: Integration of features into the customer network with laboratory and in-the-field experiments.
- Research engineer at Ericsson Research - 2012: Preparation of the 4G+ prototype for the Mobile World Congress (MWC).
- Software Developer at 4G/LTE - 2008/2011: Development in C of algorithms from standard specifications (3GPP). Extension and support of an Erlang test framework used by 50 engineers, including training sessions in Stockholm and Beijing.

Education

University of Glasgow

Glasgow, United Kingdom

PHD IN COMPUTING SCIENCE

2019

Thesis title: A Computational Approach to Gestural Interactions of the Upper Limb on Planar Surfaces.

KTH (Royal Institute of Technology)

Stockholm, Sweden

MSC IN APPLIED PHYSICS

2009

Telecom Physique Strasbourg

Illkirch-Graffenstaden, France

MSC IN ENGINEERING AND PHYSICS

2008